

NOVA HOME APPLIANCES, INC.

# Supplier & Warranty Performance Report

*Quality Analytics, Service Cost Analysis & Supplier Evaluation*

Fiscal Year 2024 | Quality & Supply Chain Division | Document Version 2.4

# 1. Warranty Performance Overview

Warranty management represents one of the most operationally complex and financially significant post-sale activities within Nova Home Appliances' value chain. Unlike manufacturing costs or logistics expenditures, warranty obligations arise asynchronously and unpredictably throughout a product's service life — which for major appliances such as refrigerators and washing machines can span a decade or more. The cost and complexity of honoring warranty commitments are therefore not merely a reflection of current manufacturing quality, but also an artifact of product and process decisions made years or even decades prior.

In FY 2024, Nova processed a total of 1,451 warranty claims across all active and legacy product categories — a reduction of 22.8% from the 1,880 claims processed in FY 2022, representing the most significant two-year improvement in claims volume since the company's quality transformation program was initiated in 2018. Total warranty service expenditure reached \$4.201 million USD in FY 2024, with an average cost per resolved claim of \$342, ranging from \$158 for Small Appliance claims to \$521 for Refrigerator claims. Field technician labor hours totaled approximately 104,000 hours across the year, serviced through Nova's network of 312 authorized service centers and 89 in-home service technicians.

The declining trend in warranty claims, visualized across eight consecutive quarters in Figure 5 of the companion Product Knowledge Base document, is a direct outcome of three parallel quality improvement streams: enhanced incoming component inspection protocols introduced in Q2 2022; the replacement of three underperforming Tier B suppliers with higher-rated Tier A alternatives in early 2023; and the deployment of SmartConnect remote diagnostics capabilities that enable proactive identification and resolution of software and sensor-related faults before they generate a physical service call. The remote diagnostics channel alone is estimated to have deflected approximately 180 potential physical warranty claim events in FY 2024.

## 1.1 Warranty Claims by Fault Type

Understanding the distribution of warranty claims by fault type is essential for directing quality improvement investment to the highest-impact areas. The table below presents the FY 2024 fault type breakdown, aggregated across all product categories. Compressor failures represent the single largest fault category by claim count at 87 claims, and also carry the highest average resolution cost due to the parts and labor intensity of compressor replacement procedures. Control board faults rank second with 74 claims, though their average resolution cost is substantially lower where remote firmware updates can substitute for physical board replacements — an increasingly feasible resolution pathway as SmartConnect penetration in the installed base grows.

| Fault Type | Claim Count | Primary Categories | Severity | Typical Resolution |
|------------|-------------|--------------------|----------|--------------------|
|------------|-------------|--------------------|----------|--------------------|

|                           |    |                            |        |                                    |
|---------------------------|----|----------------------------|--------|------------------------------------|
| Compressor Failure        | 87 | Refrigerators, AC          | High   | Compressor Replacement             |
| Control Board Fault       | 74 | All Categories             | Medium | Board Swap / Firmware Update       |
| Motor Failure             | 61 | Washing Machines           | High   | Motor Assembly Replacement         |
| Seal Leak                 | 48 | Refrigerators, Dishwashers | Low    | Seal Kit Replacement               |
| Display Error             | 42 | All Categories             | Low    | Remote Diagnostics / Board Swap    |
| Temperature Inconsistency | 39 | Refrigerators, AC          | Medium | Sensor Calibration / Replacement   |
| Door Hinge Failure        | 31 | Refrigerators, Dishwashers | Low    | Hinge Kit Replacement              |
| Noise Issue               | 28 | Washing Machines, AC       | Low    | Balance Adjustment / Pad Kit       |
| Power Failure             | 22 | All Categories             | High   | Wiring Harness / Board Replacement |
| Software Bug              | 19 | Smart-Connected Models     | Low    | OTA Firmware Push                  |

Table 1: Warranty Claims by Fault Type — FY 2024 (All Product Categories)

## 1.2 Service Cost Analysis by Category

The warranty service cost distribution across product categories reflects both the volume of claims in each category and the inherent complexity and parts cost associated with each product type. Refrigerators, as the largest and most mechanically complex product category, account for the highest total warranty spend at \$1.24 million, representing 29.6% of total warranty expenditure on 22.1% of total claims — a ratio that reflects the high labor and parts intensity of refrigeration service events relative to other categories. Small Appliances, conversely, carry the lowest per-claim cost at \$158 on average due to the relative simplicity of fault diagnosis and the prevalence of unit-exchange resolutions (where the entire product is replaced) over labor-intensive field repair.

| Category           | Total Service Cost | Cost per SKU        | Avg Cost/Claim   | Avg Part Cost    | Total Labor Hours |
|--------------------|--------------------|---------------------|------------------|------------------|-------------------|
| Refrigerators      | \$1,241,800        | \$14,273            | \$521            | \$680            | 28K hrs           |
| Washing Machines   | \$887,200          | \$14,543            | \$389            | \$520            | 21K hrs           |
| Air Conditioners   | \$743,600          | \$46,475            | \$412            | \$590            | 19K hrs           |
| Dishwashers        | \$521,400          | \$33,638            | \$298            | \$480            | 14K hrs           |
| Ovens & Cooktops   | \$487,100          | \$17,397            | \$274            | \$440            | 13K hrs           |
| Small Appliances   | \$319,900          | \$27,609            | \$158            | \$320            | 9K hrs            |
| <b>TOTAL / AVG</b> | <b>\$4,201,000</b> | <b>\$21,000 avg</b> | <b>\$342 avg</b> | <b>\$505 avg</b> | <b>104K hrs</b>   |

Table 2: Warranty Service Cost Analysis by Product Category — FY 2024

## 2. Supplier Performance Evaluation

Nova Home Appliances sources components from a network of 23 approved suppliers across 11 countries, governed by the company's Supplier Qualification Standard (SQS-2021) which establishes minimum quality system certification requirements, financial stability thresholds, ethical sourcing compliance obligations, and ongoing performance monitoring criteria. The Supplier Performance Scorecard, updated quarterly, evaluates each supplier across four weighted dimensions: delivered quality (40%), on-time delivery reliability (30%), responsiveness to quality concerns (20%), and environmental and social governance compliance (10%).

The current approved supplier list for critical-path components — defined as components whose failure directly and immediately renders the finished product non-functional — contains 10 primary suppliers summarized in the table below. These suppliers collectively represent approximately 73% of Nova's total component procurement spend. The remaining 27% is distributed across 13 secondary suppliers covering non-critical-path components such as cosmetic trim parts, packaging materials, and consumable items.

A significant evolution in Nova's supplier strategy over the past three years has been the progressive reduction of single-source dependency for critical components. As of FY 2024, seven of the ten primary critical-path components are dual-sourced, with qualified backup suppliers capable of ramping supply within 12 weeks in the event of a primary supplier disruption. This dual-source posture was tested and validated during the Q3 2023 logistics disruption that affected multiple East Asian suppliers, during which Nova successfully redirected compressor procurement to KoreComp's secondary production facility in Vietnam with only a 9-day delay impact on finished goods output.

| Supplier                | Country     | Components      | Rating | Certifications  | Quality Score | On-Time Delivery |
|-------------------------|-------------|-----------------|--------|-----------------|---------------|------------------|
| KoreComp Electronics    | South Korea | Compressors     | A      | ISO 9001        | 4.8           | 94.2%            |
| ThermoTech Components   | Germany     | Heat Exchangers | A      | ISO 9001; 14001 | 4.9           | 97.1%            |
| PolySeal Industries     | Poland      | Door Gaskets    | B      | ISO 9001        | 4.2           | 91.8%            |
| SteelForm Manufacturing | Taiwan      | Cabinet Shells  | A      | ISO 9001; IATF  | 4.6           | 95.4%            |
| ChipLogic Semiconductor | Japan       | Control Boards  | A+     | ISO 9001; AEC   | 4.9           | 98.2%            |
| MotorDrive Systems      | China       | Motors; Pumps   | B      | ISO 9001        | 4.1           | 89.6%            |

|                         |                |                |   |               |     |       |
|-------------------------|----------------|----------------|---|---------------|-----|-------|
| GlassCraft Industries   | Czech Republic | Glass Panels   | A | ISO 9001      | 4.5 | 93.7% |
| FoamTec Insulation      | Mexico         | PU Foam        | B | ISO 9001      | 4.0 | 88.9% |
| ConnectPro Wiring       | Vietnam        | Wiring Harness | B | ISO 9001      | 3.9 | 87.4% |
| FilterMaster Industries | India          | Air Filters    | A | ISO 9001; NSF | 4.4 | 92.1% |

Table 3: Critical-Path Supplier Performance Summary — FY 2024

Warranty Claims per Quarter (2023-2024)

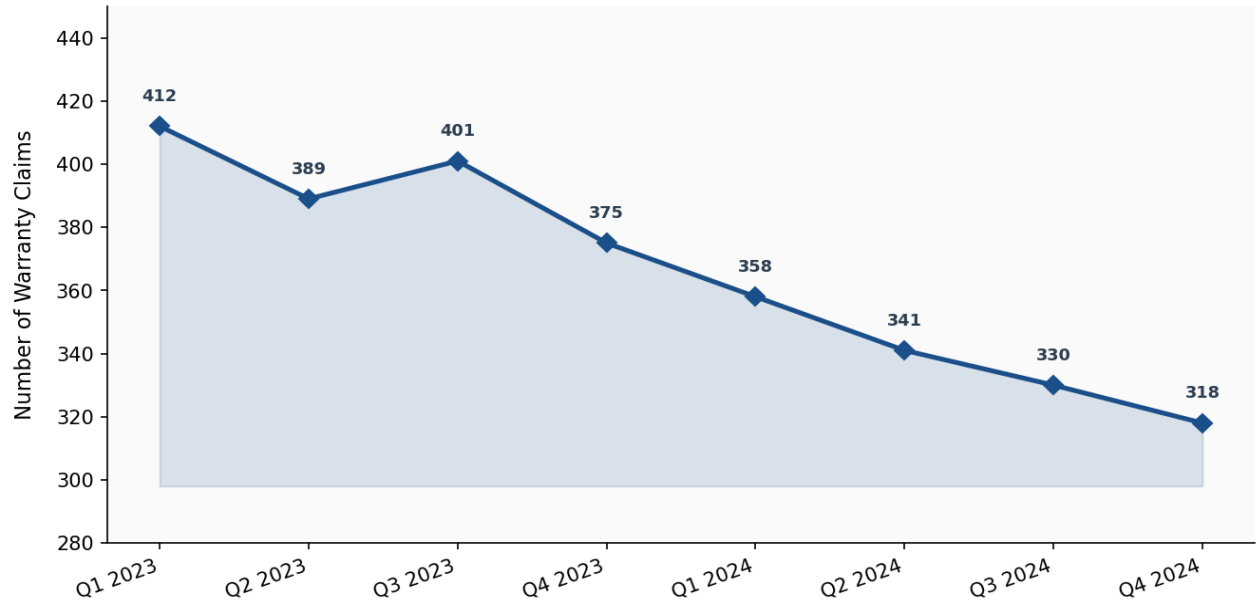


Figure 1: Declining Warranty Claims Trend — Q1 2023 to Q4 2024 (All Categories)

Customer Satisfaction by Product Category — 2024 Survey

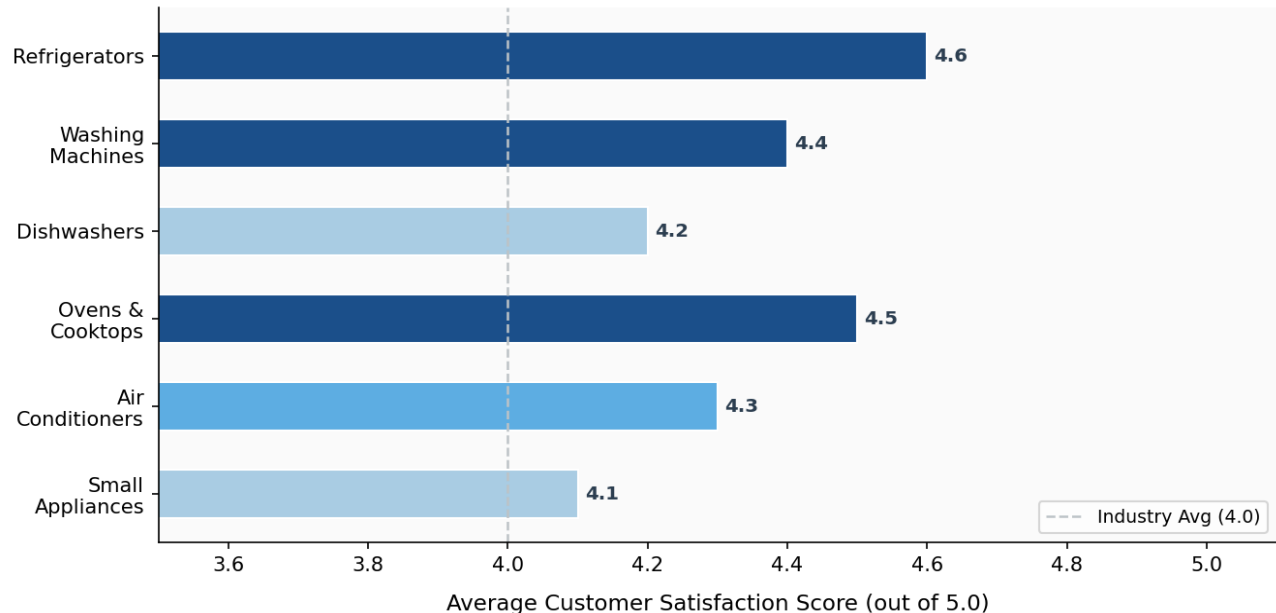


Figure 2: Customer Satisfaction Scores by Category — 2024 Survey (Industry Benchmark: 4.0)

### 2.1 Supplier Risk Assessment

Despite strong aggregate performance across the supplier network, several areas of concentration risk warrant ongoing monitoring. ConnectPro Wiring (Vietnam) carries the lowest on-time delivery score at 87.4% and has been placed on a formal Corrective Action Plan (CAP) since Q2 2024 following three consecutive quarters of delivery shortfalls exceeding the contractual threshold of 90%. The CAP requires ConnectPro to demonstrate sustained on-time performance above 91% for three consecutive months as a condition of maintaining its approved supplier status. A qualified alternative wiring harness supplier (TechWire Co., Thailand) has been identified and is currently undergoing the Nova SQS-2021 qualification audit, with completion expected in Q1 2025.

FoamTec Insulation's quality score of 4.0 and on-time delivery of 88.9% reflect intermittent raw material supply issues associated with MDI (methylene diphenyl diisocyanate) procurement volatility — a market-wide challenge affecting polyurethane foam manufacturers globally. Nova's procurement team has approved a temporary volume increase allocation to ThermoTec (Germany) for alternative insulation panel sourcing as a buffer measure while FoamTec works through its raw material stabilization program. The total dual-sourcing cost premium associated with this arrangement is estimated at \$127,000 annually, which Nova's cost-benefit analysis has determined is justified by the supply continuity risk mitigation.

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— End of Report —

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